

This notebook builds a binary classification machine learning model to predict whether a user converts (Converted = 1) or not, using user behavior and demographic data.

1. Data Understanding & Preparation: The dataset contains user-level features such as Age, Income, City Tier, Pages Viewed, Time on Site. Few of the **nominal categorical features** are Device Type and Traffic Source. The target variable is 'Converted'. Categorical variables are handled using one-hot encoding via `pandas.get_dummies()`, converting them into numeric binary columns to make them usable for the machine learning model.
2. Exploratory Data Analysis (EDA): EDA is performed to understand distributions and missing values in the variables Age, Income and Time_On_Site. Age and Time_On_Site are right and left skewed i.e., mostly the users are young people and they spend large durations on the site. Income is relatively symmetric. This step ensures informed preprocessing decisions.
3. Missing Value Handling: Simple imputation has been used- median for Age and Time_On_Site (robust to outliers) and mean for Income (better for normally distributed data). This ensures no missing values remain, which is required for training the data using the scikitlearn library.
4. Feature Scaling: Standard scaling has been applied to normalize features (mean=0, variance=1) except User_ID. This is important for Logistic Regression to ensure all features contribute equally and improve convergence.
5. Model Choice Logistic Regression: Using logistic regression for binary classification of Conversion (takes 0 or 1) works pretty as a baseline model. Class weights are set to 'balanced' to handle class imbalance, ensuring minority class is not under-represented.
6. Model Training: The model is trained using scaled features (mean= 0, SD= 1) and target labels. Training accuracy ~67.5% indicates moderate predictive ability.
7. Evaluation Metric: F1-score (~0.57) is used, balancing precision and recall, making it suitable for imbalanced classification problems.
8. Conclusion: The pipeline follows a standard ML workflow: preprocessing, encoding, scaling, modeling, evaluation. Logistic Regression serves as a strong baseline, and further improvements could include advanced models (Random Forest, KNNImputer etc.).